PIYUSH JANGDE | 22AG10034

B.Tech.(Hons.) in AGRICULTURAL AND FOOD ENGINEERING



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2026	B.TECH	IIT Kharagpur	7.04 / 10
2021	Central Board of Secondary Education	Delhi Public School, Raipur	92.2%
2019	Central Board of Secondary Education	Delhi Public School, Raipur	94.8%

PROJECTS

White Board App | React.js | Rough.js | Tailwind CSS | Vercel
- Developed a whiteboard web application to simulate real-time drawing interactions using **React** and **HTML5 Canvas**
- Utilized **Rough.js**, **Perfect Freehand** and **Tailwind CSS** for UI styling, and deployed the app on **Vercel** for live access
- Integrated tools like brush, text, shapes, eraser, stroke/fill color pickers, adjustable sizes, **undo/redo**, and **image export**
- Leveraged **React Context API** and **Reducers** for **global state management**, **useRef** for canvas handling, and **useEffect**, **useLayoutEffect** for rendering updates, side-effect control across dynamic drawing tools and responsive UI changes

Movie Tracker Web App | HTML | CSS | JavaScript | OMDb API
- Built a responsive, single-page web application to search and bookmark favorite movies using the **OMDb API**
- Implemented **asynchronous API** calls using **fetch** to retrieve movie data (title, poster, rating, plot) based on user input
- Created a **dynamic UI** with **JavaScript** to display real-time search results and maintain a personalized "Favorites" list
- Used **local storage** and interactive **DOM manipulation** to persist and update user-selected favorite movies across sessions, enabling intuitive actions like add/remove favorites, view detailed info, filter results, and reset list

Chrome Extension | HTML | CSS | JavaScript | API
- Built a Chrome extension using **HTML**, **CSS**, **JavaScript**, and **Chrome API** to bookmark problems from any page
- Used Chrome **sync storage** to make bookmarks available and consistent across all signed-in Chrome devices
- Designed a lightweight and **intuitive UI** for seamless user experience and efficient bookmark organization
- Enabled users to view and manage bookmarks regardless of the webpage they are currently browsing

Netflix-Clone | HTML | CSS
- **User Interface Design:** Designed and developed a Netflix clone, replicating the service's interface using **HTML** and **CSS**
- **Responsive Design:** Created a responsive and visually appealing design to enhance user experience across various devices
- **Front-End Development:** Integrated essential **front-end** features, like navigation, content display, and interactive elements

Solar Tracking System | Course Project | IIT Kharagpur
- **Collaborated** with a team to develop an **Arduino-based** solar tracking system aimed at maximising solar panel efficiency
- **Circuit Design:** Utilized **TinkerCAD** for designing the efficient circuitry, ensuring a robust and scalable circuit layout
- **Software Development:** Wrote and tested code in **Arduino IDE** to control the movement of the solar panels based on **LDR**
- **Prototype Creation:** Assembled a functional small-scale prototype incorporating **Arduino**, **LDR sensors** and solar panels
- **Research and Analysis:** Conducted extensive market research to propose a **cost-effective** and **energy-efficient** model

SKILLS AND EXPERTISE

Programming Languages: Proficient: C, C++, C#, JavaScript | Familiar: Python, Arduino
Software/Tools: Proficient: Visual Studio, Unity | Familiar: FreeCAD, Git, SolidWorks, TinkerCAD
Design Tools: Adobe Photoshop, Adobe Illustrator, Canva

COURSEWORK INFORMATION

Programming and Data Structures (Theory and Lab) | Probability & Statistics | Economics | Linear Algebra, Numerical and Complex Analysis | Theory of Partial Differential Equations | Advanced Calculus

POSITIONS OF RESPONSIBILITY

Senior Reporter | Awaaz | IIT Kharagpur **[Sep 2023 - Jun 2024]**
- Recruited and co-led a team of **30** juniors to organise various cultural events, enhancing campus life and student engagement
- Conducted multiple literary workshops, providing a platform for students to develop and showcase their literary talents
- Interviewed notable personalities and authored compelling articles and blogs, reaching an audience of over **10,000** on campus
- Co-authored and distributed more than **4,500** copies of the offline newsletter, addressing significant community grievances
- Secured sponsorship worth **30,000 INR** for Panch Parmeshwar, leading to a record-breaking audience of over **500** attendees

EXTRA CURRICULAR ACTIVITIES

Volunteer | National Service Scheme | IIT Kharagpur
- Trained in **CPR** and conducted discussions on critical issues such as mental health, sex education, and climate change
- Volunteered eye checkups and health checkups of villagers and had First Aid Training at the **NSS Annual Winter Camp 2023**
- Participated in the IIT Kharagpur Campus Tree Initiative '24, conducting a tree survey and collecting environmental data

Contingent | InterIIT Social and Cultural Meet 6.0
- Represented IIT Kharagpur as a member of the Scrabble contingent at the **InterIIT Social and Cultural Meet 6.0 & 7.0**
- Achieved a commendable **4th** position in the highly competitive **InterIIT** Scrabble competition